FLAT PULLBACK COMMUTES WITH CYCLE-LEVEL INTERSECTION PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B0Y: sources/stacks-chow/tags.txt, source lines 13876–13876. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a flat morphism of nonsingular varieties. Let α be a r-cycle on Y and β an s-cycle on Y. Assume that α and β intersect properly. Then $f^*\alpha$ and $f^*\beta$ intersect properly and $f^*(\alpha \cdot \beta) = f^*\alpha \cdot f^*\beta$.*

Proof. By linearity we may assume that $\alpha = [V]$ and $\beta = [W]$ for some closed subvarieties $V, W \subset Y$ of dimension r, s. Say f has relative dimension e. Then the lemma is a special case of Lemma 0B0Y because $[V] = [\mathcal{O}_V]_r$, $[W] = [\mathcal{O}_W]_r$, $f^*[V] = [f^{-1}(V)]_{r+e} = [f^*\mathcal{O}_V]_{r+e}$, and $f^*[W] = [f^{-1}(W)]_{s+e} = [f^*\mathcal{O}_W]_{s+e}$. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.